



Project WiCCED Water Quality Research Training Manual

Using uafR, Reproducible R, and Machine Learning to Understand
Coastal Water Quality

An 8-week applied undergraduate research curriculum

Program Promise

Students finish with a reproducible water-quality research project, an evidence-backed chemical and environmental data audit, at least one defensible machine-learning model, publication-style figures, a written report, a presentation, and a handoff package that can support continued Project WiCCED work.

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DSU dsDNA Core and DSU ONE Health Laboratory

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1 Program Overview

This manual is an 8-week applied research program for undergraduate interns working in the Project WiCCED Salt project space. The central training goal is practical: students learn how to turn water-quality observations, chemical screening evidence, public database records, and machine-learning models into reproducible research products that can be inspected, challenged, improved, and handed off.

The curriculum is designed for co-advising by Dr. Chase Stratton and Dr. Gulnihal Ozbay. It connects uafR, cheminformatics, reproducible R workflows, water-quality field and lab interpretation, aquaculture and coastal ecosystem context, and machine learning. It does not treat machine learning as a decoration. Models are introduced only after students understand the variables, sampling context, missingness, QA/QC flags, and evidence limits.

Program Promise

By the end of eight weeks, each student should have a complete water-quality research package: a feasible research question, a reproducible R project folder, a cleaned water-quality dataset or case dataset, documented QA/QC decisions, optional uafR chemical-screening evidence, an analysis-ready feature table, at least one defensible baseline machine-learning model, model evaluation outputs, source-backed figures and tables, a manuscript-style report, a presentation, and a continuation note.

1.1 Audience and Prerequisites

The manual assumes students are beginning undergraduate researchers. Prior experience in R, chemistry, water quality, or machine learning is helpful but not required. The program does assume sustained effort, careful documentation, and willingness to revise work after evidence checks.

Minimum technical prerequisites are:

- R and RStudio installed and working.
- Access to the uafR package from the provided student bundle or local repository.
- Ability to open CSV files, create folders, run R scripts, and read console messages carefully.
- Access to either real project data approved for training use or the teaching data included with this manual.

Minimum research prerequisites are:

- A research log that records decisions, file names, dates, errors, corrections, and interpretation limits.
- A clear distinction between observations, processed variables, model outputs, and scientific claims.

- A willingness to say “not supported by the current evidence” when the data do not justify a claim.

1.2 How to Use This Manual

The manual is organized as a working program, not a reference book. The 56-day spine should be followed in order unless the co-advising team deliberately adjusts pacing for a specific project. Weekdays are full internship days. Weekend days are lighter required asynchronous work: reading, annotation, reruns, research-log updates, peer review, project maintenance, or revision.

The curriculum has four simultaneous tracks:

- **Water-quality science:** salinity, conductivity, dissolved oxygen, nutrients, pH, temperature, turbidity, chlorophyll, microbial indicators, contaminants, field metadata, QA/QC, and ecological interpretation.
- **uafR and chemical evidence:** tentative GC-MS annotation, public database enrichment, source-backed traits, evidence trails, and chemical grouping.
- **Machine learning:** feature engineering, train/test design, baseline models, classification, regression, model diagnostics, uncertainty, interpretability, and leakage control.
- **Research communication:** figures, tables, claim-evidence-limitation writing, reproducibility statements, presentations, and handoff notes.

1.3 What Students Produce

Every student product should be saved inside a reproducible project folder. The final folder should contain:

- Project brief and question-scope record.
- Raw-data inventory and data-use note.
- Cleaned water-quality dataset or approved case dataset.
- Water-quality variable dictionary.
- QA/QC flag log.
- Optional uafR chemical-screening result object and evidence map.
- Feature table used for modeling.
- Train/test split record.
- Machine-learning model card.
- Model metrics and diagnostic plots.
- At least three publication-style figures or tables.
- Manuscript-style report.
- Presentation or poster.
- Daily research log and weekly review notes.
- Final package audit and continuation note.

1.4 What Mentors Review

The review process is evidence-centered. Mentors review whether the student can explain what each file is, why each processing step occurred, what evidence supports each variable, what the model can and cannot claim, and whether the final package can be rerun.

Review checkpoints are built into every week:

- **Week 1:** research question, project scope, and risk note.
- **Week 2:** reproducible folder and clean-session rerun.
- **Week 3:** water-quality data dictionary and QA/QC plan.
- **Week 4:** uafR evidence map or chemical-screening decision note.
- **Week 5:** first defensible baseline model and validation plan.
- **Week 6:** feature table, leakage audit, and model comparison.
- **Week 7:** evidence-based interpretation and figure package.
- **Week 8:** final report, presentation, archive, and continuation note.

1.5 Scientific Integrity Rules

Claims That Must Stay Conservative

This program trains students to interpret evidence, not to decorate uncertain results. A tentative GC-MS match is not a confirmed chemical identity. A PubChem or KEGG record is not evidence that a compound is present in a water sample. A pathway annotation is not evidence of pathway activity. A hazard label is not a site-specific risk assessment. A machine-learning model association is not proof of causation. A high-accuracy model can still be invalid if it leaks site, time, treatment, or post-outcome information.

Acceptable wording is specific and limited. For example:

- Stronger: “The tentative GC-MS feature matched caffeine in the library search and uafR linked the query to PubChem CID 2519; this supports follow-up confirmation, not confirmed sample identity.”
- Stronger: “Conductivity, salinity, temperature, and site context were useful predictors in the teaching model under the recorded split; external validation is required before operational use.”
- Weaker: “The site contains caffeine and is biologically active.”
- Weaker: “The model proves saltwater intrusion is causing the observed biological response.”

1.6 Weekly Time Budget

A typical weekday contains 6 to 7 active hours:

- 30 minutes: check-in, daily objective, research-log setup.
- 60 to 90 minutes: concept block or reading discussion.

- 90 to 120 minutes: guided technical lab.
- 90 to 150 minutes: independent project work.
- 45 to 60 minutes: evidence, writing, figure, or documentation block.
- 30 minutes: peer or mentor checkpoint.

A typical weekend day contains 1 to 3 focused hours. Weekend work should not introduce major new methods unless the student is already ahead. It should consolidate the week: annotate a reading, rerun scripts, revise a worksheet, clean the research log, prepare questions, or review a peer project.

1.7 Skill Progression

The program moves students through four stages:

- **Operator:** runs scripts, checks objects, records errors, and follows data-handling rules.
- **Inspector:** examines missingness, suspicious values, duplicate records, source coverage, and model diagnostics.
- **Analyst:** chooses variables, compares models, creates figures, and explains uncertainty.
- **Researcher:** writes defensible claims, connects evidence to literature, and hands off a reproducible project.

1.8 Before Week 1

Before the first day, the project lead should prepare:

- The student bundle or local uafR installation path.
- This manual as a PDF and source folder.
- The `training_wicced_water_quality/scripts` folder.
- Any approved real water-quality or GC-MS data that students may use.
- A decision about whether students will work with real data, teaching data, or both.
- A data privacy note for unpublished or partner-owned datasets.
- A shared schedule for weekly review checkpoints.

Students should arrive with RStudio installed, a folder location selected for project work, and a research log ready to begin. If installation is incomplete, Week 1 can still proceed using paper worksheets, source readings, and the teaching data dictionary.

2 Project WiCCED Water-Quality Context

Project WiCCED Salt focuses on water in the complex coastal environment of Delaware. The project space includes saltwater intrusion, freshwater-resource vulnerability, drinking water, farmland, coastal ecosystems, infrastructure, wetland migration, aquaculture, stakeholder decision-making, and workforce development. The student training program therefore treats water-quality data as both environmental evidence and decision-relevant information.

2.1 Project Space

The WiCCED project emphasizes saltwater intrusion as a serious risk to Delaware's drinking water, farmland, coastal ecosystems, and infrastructure. The research plan spans urban and suburban systems, rural systems, and natural systems. It includes physical-risk mapping, water and soil salinization, groundwater and surface-water context, stormwater and wastewater infrastructure, salinity effects on organisms, salt-tolerant crops, aquaculture and benthic fauna, marsh migration, water-quality ecosystem services, and stakeholder response.

This manual translates that project space into student-scale research. A student project should be narrower than the full WiCCED portfolio. Good student projects ask questions such as:

- Which measured variables best separate higher-salinity and lower-salinity sampling contexts in a teaching or project dataset?
- How do missingness, season, site, and method affect the apparent relationship between salinity and dissolved oxygen?
- Can a simple model predict whether a sample belongs to a high-salinity class using only variables available before interpretation?
- Do tentative organic chemical features from GC-MS align with source-backed chemical traits that could guide follow-up sampling?
- Which sites, dates, or sample groups require the most caution because source coverage, missing data, or model uncertainty is high?

2.2 Water-Quality Variables

Water quality is not a single measurement. It is a structured set of chemical, physical, biological, spatial, temporal, and methodological observations. A WiCCED-aligned student should be able to define each variable before modeling it.

Variable family	Research use	Interpretation caution
Salinity, conductivity, chloride	Directly relevant to saltwater intrusion, mixing gradients, aquaculture stress, and infrastructure exposure.	Instrument calibration, tidal stage, depth, season, rainfall, and ground-water influence can change interpretation.
Temperature	Controls oxygen solubility, biological activity, and seasonal structure.	Temperature can be a confounder in almost every water-quality model.
Dissolved oxygen	Important for aquatic-life stress and ecosystem condition.	Low dissolved oxygen can reflect temperature, stratification, organic matter, respiration, time of day, or measurement issues.
pH and alkalinity	Relevant to aquatic chemistry, organism stress, and carbon-system interpretation.	pH readings depend on calibration, temperature compensation, and sample handling.
Turbidity and clarity	Useful for sediment, runoff, algal conditions, and visual water quality.	High turbidity does not identify the source of particles by itself.
Nitrogen and phosphorus species	Link to nutrient enrichment, eutrophication, runoff, and management questions.	Units, detection limits, filtered/unfiltered status, and analytical method must be recorded.
Chlorophyll and algal indicators	Useful for productivity and bloom-related questions.	Pigment readings are indirect and can vary by method, community composition, and degradation.
Bacteria and biological indicators	Relevant to human use, shellfish areas, and ecosystem health.	Single measurements can be episodic; sampling conditions and holding time matter.
Organic chemical features	Can support contaminant screening, source tracing, or organic-matter interpretation when analyzed carefully.	Tentative GC-MS matches are not confirmed identities without standards and appropriate confirmation.
Spatial and land-context variables	Connect water data to watershed, marsh, aquaculture, agriculture, infrastructure, and land-use questions.	Spatial proxies can leak site identity into models and can be mistaken for causal mechanisms.

2.3 How uafR Fits

uafR is useful when the project includes GC-MS chemical screening or chemical-name lists that require structured source evidence. In this manual, uafR is not presented as a replacement for water-quality sensors, nutrient chemistry, microbiology, or field QA/QC. Instead, uafR supports a specific layer of the project:

- Processing GC-MS hit tables with `spreadOut()`, `mzExacto()`, and `exactoThese()`.
- Enriching chemical query lists with `categorate(detail = "research")`.

- Auditing source coverage with `validateCategorateResult()` and validation tables.
- Converting source-backed annotations into analysis-ready traits with `chemicalTraitMatrix()` and `chemicalTraitEvidence()`.
- Using `pubchemProfile()` for targeted PubChem profiles when a small property-focused query is more appropriate than a full categorate run.

uafR Boundary

uafR can help summarize tentative chemical evidence and public database records. It cannot confirm the presence of a compound in a water sample, quantify exposure, identify toxicity at a site, or replace field/lab water-quality methods. Confirmed chemical identity requires appropriate analytical confirmation, standards, retention-time evidence, spectra, and method validation.

2.4 How Machine Learning Fits

Machine learning is useful in this program when it answers a question that cannot be handled well by simple inspection alone. It may help predict a water-quality indicator, classify samples into interpretable status groups, identify unusual observations, compare feature sets, or support prioritization for follow-up sampling.

Machine learning is not useful if the variable definitions are unclear, the data are too small for the model being used, the split leaks future or site identity, the model is not compared to a simple baseline, or the output is interpreted as causal proof. Every model in this program must have:

- A clearly defined prediction target.
- A feature table created before the target is interpreted.
- A split plan that protects against leakage.
- A baseline model.
- Metrics appropriate to the target.
- A model card documenting data, features, performance, uncertainty, and limits.

2.5 Co-Advised Research Identity

The co-advised training space is intentionally interdisciplinary. Dr. Ozbay's expertise connects the program to water resources, aquaculture, aquatic ecology and health, habitat restoration, Eastern oyster and blue crab systems, and nutrient/runoff stressors. Dr. Stratton's expertise connects the program to data science, uafR, cheminformatics, chemical ecology, coding, artificial intelligence, and machine learning. Students should learn to keep both sides visible: environmental meaning and computational rigor.

2.6 Student-Safe Claim Templates

Evidence type		Can support	Cannot support by itself
Water-quality sensor or lab value		A measured condition at a time, place, depth, and method.	Long-term trend, regulatory conclusion, or mechanism without design support.
Tentative match	GC-MS	A candidate chemical feature for follow-up and source-backed annotation.	Confirmed identity or concentration.
PubChem or KEGG annotation		Known public database information about a compound record.	Presence, activity, toxicity, or pathway function in the sample.
uafR trait matrix		A reproducible source-backed feature table for grouping or modeling.	Biological mechanism unless validated by independent evidence.
Machine-learning prediction	pre-	Pattern learned under the recorded training/test design.	Causation, field deployment readiness, or generalization beyond tested data.
Literature match		A peer-reviewed comparison point.	Proof that the same mechanism applies to the student's dataset.

Recommended wording:

- “The data are consistent with a salinity gradient under the measured conditions.”
- “The model prioritized conductivity and temperature under this split; these variables should be interpreted as predictive features, not confirmed causal drivers.”
- “The uafR output provides a source-backed trait for the queried compound; the sample-level presence of that compound remains tentative.”
- “This result motivates targeted confirmation rather than final environmental interpretation.”

3 Eight-Week Training Spine

The training spine covers 56 active calendar days. Weekdays are full internship days. Weekend days are lighter required asynchronous work. Each day produces a concrete artifact or revision that moves the final project forward.

3.1 Week 1: WiCCED Onboarding and Question Design

Week 1: WiCCED onboarding and water-quality question design

Project milestone: Students understand the project space, establish research-log habits, and select a feasible water-quality question.

Primary product: One-page project brief, chemical/water-quality intake table, first risk note, and Week 1 research-log entries.

Main tool emphasis: WiCCED context, water-quality variables, responsible conduct, research logs, and scope control.

Day 1: Orientation to WiCCED, dsDNA Core, and ONE Health Water Research

Time design: Full day: 30 min check-in; 75 min project-space briefing; 90 min source walkthrough; 120 min research-log setup; 60 min writing; 30 min checkpoint.

Purpose: Begin the program with a concrete understanding of saltwater intrusion, Delaware water quality, coastal ecosystems, aquaculture connections, and reproducible research expectations.

Tools and materials: This manual, Project WiCCED website notes, research log template, project folder, and source bibliography.

Work blocks: Read the program overview; build the first research-log entry; summarize why saltwater intrusion matters for water quality, aquaculture, agriculture, wetlands, and communities; list three possible student-scale questions.

End-of-day product: Research log opened, folder created, and three candidate questions written with one sentence of motivation each.

Quality gate: The log includes date, file location, sources consulted, and at least one uncertainty that needs review.

Day 2: Turning Broad Interests into Testable Questions

Time design: Full day: 30 min objective; 90 min question-design clinic; 105 min worksheet lab; 120 min independent revision; 45 min evidence writing; 30 min review.

Purpose: Convert a broad interest such as oyster health, salinity intrusion, nutrients, contaminants, or marsh water quality into a feasible 8-week research question.

Tools and materials: Project Brief worksheet, Water-Quality Variable Dictionary worksheet, WiCCED context chapter, and optional literature search notes.

Work blocks: Draft candidate questions; identify response variables, predictors, sample units, and evidence limits; reject questions that require unavailable data or causal claims beyond the design.

End-of-day product: Two refined candidate questions with variables, data needs, and feasibility notes.

Quality gate: Each question can be answered with available or teaching data and has a clear product by Week 8.

Day 3: Chemical Identity, Water-Quality Evidence, and Tentative Annotation

Time design: Full day: 30 min check-in; 90 min concept block; 90 min uafR boundary discussion; 120 min evidence-type sorting; 60 min log update; 30 min checkpoint.

Purpose: Learn why chemical screening evidence, sensor data, nutrient measurements, and database annotations have different claim strength.

Tools and materials: uafR function docs, claim-template table, GC-MS annotation notes, and research log.

Work blocks: Sort example claims into measured observation, tentative annotation, public database descriptor, model output, and confirmed evidence; write stronger wording for weak claims.

End-of-day product: Claim-evidence-limitation table for at least eight example statements.

Quality gate: No statement treats a tentative match, database record, or model prediction as confirmed sample truth.

Day 4: Evidence Source Landscape for WiCCED Projects

Time design: Full day: 30 min objective; 90 min source briefing; 90 min database/literature audit; 120 min project-specific source map; 60 min writing.

Purpose: Understand what PubChem, KEGG, LOTUS, FEMA, FDA/SPL, MeSH, DNREC, literature, and project datasets can and cannot support.

Tools and materials: Database Evidence Map worksheet, web source notes, uafR documentation, and bibliography.

Work blocks: Map at least five source types to the student's project question; identify whether each source provides measurements, annotations, identifiers, pathways, occurrence, regulatory context, or literature.

End-of-day product: Database Evidence Map v1.

Quality gate: Every source entry has a use, a limitation, and a citation or file path.

Day 5: Scope Gate and Project Selection

Time design: Full day: 30 min check-in; 60 min project pitch prep; 90 min peer review; 90 min mentor review; 120 min revision; 45 min readiness checklist.

Purpose: Select one primary project question and define the minimum viable project.

Tools and materials: Project Brief worksheet, evidence map, risk note, file inventory, and research log.

Work blocks: Present the project question; revise scope after review; define required variables, optional uafR chemical evidence, expected figures, and ML target.

End-of-day product: Approved Project Brief v1 and Week 2 readiness checklist.

Quality gate: The project has a defined sample unit, target outcome or comparison, data source, risks, and fallback teaching-data path.

Day 6: Weekend Reading Annotation and Intake Revision

Time design: Weekend: 1 to 2.5 hours.

Purpose: Consolidate Week 1 by annotating one source and revising the intake table.

Tools and materials: One selected reading or official project source, Chemical/Water-Quality Intake Table worksheet, and research log.

Work blocks: Annotate the source; add variables, chemical names, sites, organisms, or methods that may matter; revise the intake table.

End-of-day product: Annotated reading note and updated intake table.

Quality gate: The note separates useful facts from questions that still need verification.

Day 7: Weekend Reflection and Week 2 Readiness

Time design: Weekend: 1 to 2 hours.

Purpose: Prepare to move from question design to reproducible project construction.

Tools and materials: Research log, Project Brief v1, evidence map, and readiness checklist.

Work blocks: Write a one-page reflection on what changed during Week 1; list installation, data, and file-organization issues to solve in Week 2.

End-of-day product: Week 1 reflection and Week 2 action list.

Quality gate: A peer or mentor can understand the project question and next steps from the project folder alone.

3.2 Week 2: R, RStudio, Git, and Reproducible Project Structure

Week 2: Reproducible water-quality project build

Project milestone: Students build a clean RStudio project that can be rerun from the project root.

Primary product: R project skeleton, README, script order, install/session notes, and clean-session rerun record.

Main tool emphasis: RStudio, base R, uafR readiness, file paths, script hygiene, Git habits, and bundle use.

Day 8: R/RStudio and uafR Readiness Check

Time design: Full day: 30 min check-in; 90 min installation audit; 90 min setup script; 120 min project skeleton; 60 min documentation.

Purpose: Make sure the student's local machine can run the training scripts and find uafR.

Tools and materials: scripts/00_setup_check.R, scripts/01_project_setup.R, RStudio, student bundle, and project folder.

Work blocks: Run the setup check; create the project skeleton; record R version, package status, and any missing dependencies.

End-of-day product: Project folder with subfolders, setup log, and session note.

Quality gate: A clean R session can run the setup check and report uafR status without hidden local paths.

Day 9: Reading, Inspecting, Cleaning, and Writing Tables in R

Time design: Full day: 30 min objective; 90 min table concepts; 120 min guided lab; 120 min independent practice; 45 min log.

Purpose: Learn how to inspect CSV tables before using them in analysis or models.

Tools and materials: Teaching water-quality CSV, `scripts/02_water_quality_qaqc.R`, data dictionary worksheet, and RStudio.

Work blocks: Read the teaching dataset; inspect dimensions, column names, types, missingness, ranges, and units; write a cleaned copy and QA/QC summary.

End-of-day product: Cleaned teaching-data copy and first QA/QC summary CSV.

Quality gate: The student can explain each column, its unit, its missingness, and one suspicious-value rule.

Day 10: Git Basics, Safe Paths, and Repository Habits

Time design: Full day: 30 min check-in; 75 min Git concept block; 90 min practical lab; 120 min README revision; 60 min peer handoff.

Purpose: Use version control habits that protect data, scripts, and reproducibility.

Tools and materials: Git or RStudio Git pane, README draft, .gitignore template, and research log.

Work blocks: Initialize or inspect a local repository; make a non-sensitive commit if Git is available; update README with script order and data rules; document what files should never be committed.

End-of-day product: README v1, .gitignore note, and commit or Git-status screenshot/note.

Quality gate: No private raw data, credentials, machine-specific cache folders, or absolute personal paths are required to rerun the project.

Day 11: Numbered Workflows and Raw Versus Processed Data

Time design: Full day: 30 min objective; 90 min workflow design; 120 min script-order lab; 90 min independent organization; 60 min writing.

Purpose: Create a script order that preserves raw data and makes processed outputs traceable.

Tools and materials: Project folder, README, teaching data, and script templates.

Work blocks: Define `data_raw`, `data_processed`, `results`, `figures`, `models`, `logs`, and `reports`; add a script index and expected outputs.

End-of-day product: Script-order table in README and updated folder tree.

Quality gate: A new user can tell which script to run first and which files are inputs versus outputs.

Day 12: Clean-Session Reproducibility Check

Time design: Full day: 30 min check-in; 90 min rerun protocol; 120 min clean-session rerun; 120 min debugging; 45 min sign-off note.

Purpose: Verify that Week 2 work can be rerun without relying on hidden console objects.

Tools and materials: RStudio restart, setup scripts, QA/QC script, README, and research log.
Work blocks: Restart R; run scripts in order; fix path errors; record failures and corrections; produce a clean-session rerun note.
End-of-day product: Clean-session rerun record and reproducibility checkpoint.
Quality gate: All Week 2 outputs can be recreated from scripts and documented inputs.

Day 13: Weekend Debugging Log and Session Information

Time design: Weekend: 1 to 2 hours.
Purpose: Turn setup issues into documented knowledge.
Tools and materials: Research log, session information, console notes, README.
Work blocks: Record R version, package versions, operating system, unresolved errors, and command recap.
End-of-day product: Weekend debugging log and session-info note.
Quality gate: The log is specific enough that another student can reproduce the issue or verify that it is fixed.

Day 14: Weekend Peer Handoff

Time design: Weekend: 1 to 2.5 hours.
Purpose: Test whether the project folder is understandable without verbal explanation.
Tools and materials: README, folder tree, setup log, and peer review form.
Work blocks: Ask a peer or mentor to inspect the folder from README alone; collect questions; revise confusing sections.
End-of-day product: Peer handoff note and README revision.
Quality gate: The reviewer can identify the project question, data location, script order, and current outputs.

3.3 Week 3: Water-Quality Data Literacy and QA/QC

Week 3: Water-quality data literacy and QA/QC

Project milestone: Students build a defensible water-quality dataset and understand field/lab metadata.
Primary product: Water-quality variable dictionary, QA/QC flag log, missingness report, and data-readiness gate.
Main tool emphasis: Units, detection limits, field metadata, method effects, missingness, outliers, and Delaware monitoring context.

Day 15: Sample Units, Sites, Dates, and Metadata

Time design: Full day: 30 min check-in; 90 min metadata concept block; 120 min data dictionary lab; 120 min project update; 45 min writing.
Purpose: Define what one row means and which metadata are required for interpretation.
Tools and materials: Teaching data, project data if available, Variable Dictionary worksheet, and QA/QC Flag Log.

Work blocks: Identify sample unit, site, date/time, depth, tide/rainfall context if available, method, units, and responsible data steward.

End-of-day product: Water-Quality Variable Dictionary v1.

Quality gate: Every variable has a unit, type, source, method note, and interpretation caveat.

Day 16: Ranges, Units, Detection Limits, and Suspicious Values

Time design: Full day: 30 min objective; 90 min QA/QC rules; 120 min script lab; 120 min project-specific flags; 60 min review.

Purpose: Create transparent rules for impossible, suspicious, missing, and censored water-quality values.

Tools and materials: scripts/02_water_quality_qaqc.R, QA/QC worksheet, teaching data, and project data if approved.

Work blocks: Run QA/QC checks; adapt range rules cautiously; flag values without deleting them; write a decision note for each rule.

End-of-day product: QA/QC flag log and flagged-data output.

Quality gate: The student can explain why each flag exists and can recover the original raw value.

Day 17: Salinity, Conductivity, Dissolved Oxygen, and Nutrients

Time design: Full day: 30 min check-in; 90 min variable interpretation; 90 min exploratory lab; 120 min independent summaries; 60 min writing.

Purpose: Understand core variables often used in saltwater intrusion and aquatic ecosystem interpretation.

Tools and materials: Teaching data, DNREC monitoring context notes, variable dictionary, and plotting script template.

Work blocks: Summarize salinity, conductivity, dissolved oxygen, pH, turbidity, nitrate, chlorophyll, and temperature by site/date/context; identify likely confounders.

End-of-day product: Exploratory summary table and two draft plots.

Quality gate: No plot or summary implies causation; all units and sample counts are visible.

Day 18: Missingness, Bias, and Sampling Design

Time design: Full day: 30 min objective; 90 min missingness concepts; 120 min missingness audit; 120 min project risk revision; 45 min checkpoint.

Purpose: Learn how missing values, uneven sampling, season, site, and method can bias analysis and ML models.

Tools and materials: Source Coverage/Missingness worksheet, teaching data, project data, and research log.

Work blocks: Compute missingness by variable, site, and date; identify structural versus accidental missingness; decide which variables are usable.

End-of-day product: Missingness report and project risk update.

Quality gate: Variables with heavy or biased missingness are either excluded or clearly marked as exploratory.

Day 19: Data-Readiness Gate

Time design: Full day: 30 min check-in; 90 min review preparation; 90 min mentor review; 120 min revision; 60 min documentation.

Purpose: Freeze the first analysis-ready water-quality dataset or approve the teaching-data fallback.

Tools and materials: Variable dictionary, QA/QC log, missingness report, README, and cleaned data.

Work blocks: Present the data pipeline; answer what each row means; revise after review; define the Week 4 chemical/uafR path.

End-of-day product: Data-readiness gate memo.

Quality gate: The memo states which dataset will be used, which variables are included/excluded, and why.

Day 20: Weekend Reading Annotation on Water Quality

Time design: Weekend: 1 to 3 hours.

Purpose: Connect data literacy to a source on Delaware water monitoring, aquaculture, marsh systems, salinity, or machine learning.

Tools and materials: Approved reading, research log, and evidence map.

Work blocks: Annotate one source; extract variables, methods, and limitations; connect it to the project dictionary.

End-of-day product: Annotated reading and updated dictionary notes.

Quality gate: The annotation records how the source changes variable interpretation or project scope.

Day 21: Weekend Data Dictionary Revision

Time design: Weekend: 1 to 2 hours.

Purpose: Clean up Week 3 artifacts before chemical screening or modeling begins.

Tools and materials: Variable dictionary, QA/QC flag log, missingness report, README.

Work blocks: Resolve inconsistent variable names; revise units; add missing caveats; prepare Week 4 questions.

End-of-day product: Data dictionary v2 and Week 4 readiness checklist.

Quality gate: A reader can understand the analysis-ready data without opening the raw file first.

3.4 Week 4: uafR Chemical Screening and Evidence Mapping

Week 4: uafR chemical evidence for water-quality questions

Project milestone: Students learn when uafR is relevant, run an offline workflow, and create a cautious chemical evidence map.

Primary product: uafR workflow output or decision note, chemical evidence map, query settings log, and claim-limit memo.

Main tool emphasis: GC-MS hit tables, tentative annotation, PubChem profiles, categorize research output, validation, and trait evidence.

Day 22: GC-MS Hit Tables and Water-Quality Chemical Screening

Time design: Full day: 30 min objective; 90 min GC-MS hit-table concept block; 120 min input audit lab; 120 min project relevance decision; 45 min writing.

Purpose: Understand what GC-MS hit tables can add to a water-quality project and when they are not needed.

Tools and materials: uafR docs, GC-MS Input Column Audit worksheet, optional project chemical data, and teaching examples.

Work blocks: Inspect required columns for `spreadOut()`; identify compound names, retention time, area, match factor, file names, and suspicious rows; decide whether chemical screening belongs in the project.

End-of-day product: GC-MS input audit or chemical-screening not-applicable note.

Quality gate: The project does not force uafR chemical analysis if no defensible chemical input exists.

Day 23: Simulated Core uafR-Compatible Workflow

Time design: Full day: 30 min check-in; 90 min function map; 120 min guided run; 120 min object inspection; 60 min log.

Purpose: Run and inspect a simulated WiCCED-themed GC-MS workflow without relying on live web services.

Tools and materials: `data/wicced_simulated_gcms_hits.csv`, simulated query/reference files, `spreadOut()`, `mzExacto()`, and `scripts/03_uafR_chemical_screening_template.R`.

Work blocks: Run the simulated pipeline; inspect the raw hit table, spread RDS, exact-match output, long abundance table, context summary, and evidence notes; record which operations mirror uafR hit-table processing and which live functions require web lookup.

End-of-day product: Simulated GC-MS workflow note and saved output tables.

Quality gate: The student can explain each required input column, why duplicate detections are aggregated, and why simulated chemical patterns are not environmental evidence.

Day 24: Research-Grade Enrichment and Validation First

Time design: Full day: 30 min objective; 90 min enrichment concept block; 90 min small-query planning; 120 min optional live/saved result inspection; 60 min validation notes.

Purpose: Learn the safe pattern for `categorate(detail = "research")` and validation outputs.

Tools and materials: uafR docs, `validateCategorateResult()`, Query Settings Log, saved result if available, and cache/throttle notes.

Work blocks: Plan a tiny query list; use saved output or run only if network and package dependencies are ready; inspect `ValidationSummary`, `SourceCoverage`, `TableQuality`, `DataDictionary`, `ChemicalTraits`, and `ChemicalTraitEvidence`.

End-of-day product: Categorate validation note and query settings log.

Quality gate: No live web query is scaled until the small run or saved example is understood and documented.

Day 25: Chemical Traits for Water-Quality Grouping

Time design: Full day: 30 min check-in; 90 min trait concepts; 120 min matrix lab; 120 min project-specific evidence mapping; 45 min review.

Purpose: Convert source-backed chemical annotations into variables that could be used for grouping or modeling.

Tools and materials: `chemicalTraitMatrix()`, `chemicalTraitEvidence()`, Trait Evidence Audit worksheet, and saved categorate result if available.

Work blocks: Inspect trait names, confidence, sources, evidence text, and missingness; build or inspect a binary/count/confidence trait matrix; reject narrative-only fields as model features unless discretized reproducibly.

End-of-day product: Trait Evidence Audit v1.

Quality gate: Every retained trait has a source table, source field, extraction rule, confidence note, and caveat.

Day 26: Claim-Limit Memo for Chemical Evidence

Time design: Full day: 30 min objective; 90 min strong/weak claims; 90 min evidence court; 120 min writing; 60 min checkpoint.

Purpose: Write a project-specific memo that prevents overclaiming from tentative chemical evidence.

Tools and materials: Evidence Court worksheet, claim templates, uafR outputs or decision note, and research log.

Work blocks: Trace five possible chemical statements back to evidence; rewrite weak claims; list confirmation work required for stronger claims.

End-of-day product: Chemical claim-limit memo.

Quality gate: The memo clearly distinguishes tentative matches, database descriptors, source-backed traits, and confirmed identities.

Day 27: Weekend Wrong-Input Troubleshooting

Time design: Weekend: 1 to 2.5 hours.

Purpose: Practice diagnosing malformed chemical-screening inputs without damaging raw data.

Tools and materials: Wrong-Column GC-MS Input Repair lab, example malformed table, and research log.

Work blocks: Identify missing or mislabeled columns; write what the error means; create a corrected copy with documented changes.

End-of-day product: Troubleshooting note and corrected example file.

Quality gate: The raw file remains unchanged and every correction is documented.

Day 28: Weekend Chemical Evidence Archive

Time design: Weekend: 1 to 2 hours.

Purpose: Prepare chemical evidence for modeling decisions in Week 5.

Tools and materials: uafR notes, query settings log, evidence map, README.

Work blocks: Archive saved outputs or write the not-applicable note; update README with chemical-evidence status.

End-of-day product: Chemical evidence archive and Week 5 readiness checklist.

Quality gate: The project folder states exactly whether uafR chemical evidence will enter the ML feature table.

3.5 Week 5: Machine-Learning Foundations for Water Quality

Week 5: Machine-learning foundations

Project milestone: Students build a baseline model from a documented feature table and evaluate it honestly.

Primary product: ML feature table v1, train/test split plan, baseline regression or classification model, and first model card.

Main tool emphasis: Prediction targets, leakage, baseline models, metrics, cross-validation concepts, and model cards.

Day 29: Prediction Questions and Target Definition

Time design: Full day: 30 min check-in; 90 min ML framing; 120 min target-design lab; 120 min feature inventory; 45 min writing.

Purpose: Define a prediction or classification target that is meaningful and available before interpretation.

Tools and materials: ML Feature Inventory worksheet, Train/Test Split Plan, teaching data, project data, Week 3 outputs, and `ml_resources/ml_glossary.csv`.

Work blocks: Choose one target such as salinity, dissolved oxygen, nitrate, or high-salinity class; list candidate predictors; remove features that duplicate or reveal the target; read the glossary terms for target, feature, split, baseline, leakage, and model card.

End-of-day product: ML Feature Inventory v1 and target definition note.

Quality gate: The target is measurable, not circular, and not created from predictors that will also be used as model inputs.

Day 30: Feature Table and Split Script

Time design: Full day: 30 min objective; 90 min leakage examples; 120 min guided script run; 120 min output inspection; 60 min review.

Purpose: Protect model evaluation from overly optimistic results by building a documented feature table and split plan.

Tools and materials: Train/Test Split Plan worksheet, teaching data, site/date metadata, `scripts/04_ml_01_build_feature_table.R`, and `ml_resources/ml_workflow_checklist.csv`.

Work blocks: Run the feature-table script; open the feature dictionary, training rows, testing rows, and split plan; identify leakage risks from site labels, dates, conductivity, post-outcome variables, duplicates, and repeated sampling.

End-of-day product: Feature table, feature dictionary, and approved train/test split plan.

Quality gate: The split matches the intended use case and every included/excluded variable has a documented reason.

Day 31: Explore Before Modeling

Time design: Full day: 30 min check-in; 90 min exploration concepts; 120 min guided script run; 120 min plot/table inspection; 45 min log.

Purpose: Inspect the feature table before fitting models.

Tools and materials: `scripts/04_ml_02_explore_features.R`, feature table, feature dictionary, exploration plots, and research log.

Work blocks: Run the exploration script; inspect feature summaries, correlations, class balance, and plots; answer the beginner exploration questions before any model interpretation.

End-of-day product: Feature summary, correlation table, class balance table, exploration plots, and answered exploration questions.

Quality gate: The student can name at least one plausible feature, one weak feature, one excluded feature, and one limitation before modeling.

Day 32: Train Regression and Classification Models

Time design: Full day: 30 min objective; 90 min regression/classification concepts; 120 min guided model run; 120 min metrics review; 60 min writing.

Purpose: Build simple baseline models and compare them to candidate models.

Tools and materials: `scripts/04_ml_03_train_models.R`, feature table, split plan, and glossary.

Work blocks: Run the training script; compare mean-only regression to linear regression; compare majority-class classification to logistic classification; inspect RMSE, MAE, confusion-matrix counts, sensitivity, specificity, and balanced accuracy.

End-of-day product: Regression metrics, classification metrics, predictions table, coefficients table, and model-comparison summary.

Quality gate: The student can explain false positives, false negatives, baseline comparison, and why test-set performance matters.

Day 33: Diagnostics, Model Card, and First ML Review

Time design: Full day: 30 min check-in; 90 min diagnostic concepts; 90 min script run; 120 min model-card review; 60 min checkpoint.

Purpose: Document the first model so its purpose, data, limits, and performance are visible.

Tools and materials: `scripts/04_ml_04_diagnostics_and_model_card.R`, `ml_resources/model_card_field_guide.md`, metrics CSVs, plots, predictions, and research log.

Work blocks: Run the diagnostics script; inspect observed-versus-predicted plots, residuals, classification probabilities, diagnostic summary, model card, and interpretation guide; revise features, split, metrics, and limitations; list next experiments for Week 6.

End-of-day product: Diagnostic plots, diagnostic summary, Model Card v1, and ML review note.

Quality gate: The model card names the target, rows used, exclusions, features, split design, metrics, diagnostics, limitations, and prohibited uses.

Day 34: Weekend ML Reading Annotation

Time design: Weekend: 1 to 3 hours.

Purpose: Connect the baseline model to water-quality ML literature without copying claims.

Tools and materials: Approved ML review or methods article, model card, research log.

Work blocks: Annotate one ML water-quality source; identify target types, algorithms, validation designs, and uncertainty concerns.

End-of-day product: Annotated ML reading and model-card update.

Quality gate: The annotation improves the student's validation plan or feature rationale.

Day 35: Weekend ML Rerun and Sensitivity Note

Time design: Weekend: 1 to 2.5 hours.

Purpose: Check whether the model result changes under a simple defensible sensitivity run.

Tools and materials: Beginner ML scripts, feature inventory, model card, and research log.

Work blocks: Rerun the ladder after removing one feature or changing one documented setting; record differences without cherry-picking.

End-of-day product: Sensitivity note.

Quality gate: The note reports what changed, whether the baseline comparison changed, and why the preferred result should remain cautious.

3.6 Week 6: Feature Engineering, Model Comparison, and Spatial/Temporal Context

Week 6: Feature engineering and model comparison

Project milestone: Students build an analysis-ready feature table, compare models, and inspect spatial/temporal context without leakage.

Primary product: Frozen feature table, model comparison table, leakage audit, and model interpretation memo.

Main tool emphasis: Feature engineering, grouping variables, chemical traits, spatial context, temporal structure, model comparison, and uncertainty.

Day 36: Feature Table Build

Time design: Full day: 30 min objective; 90 min feature-engineering rules; 120 min table build; 120 min independent revision; 60 min review.

Purpose: Create one feature table used by all Week 6 modeling.

Tools and materials: ML Feature Inventory, water-quality cleaned data, optional uafR trait matrix, and README.

Work blocks: Join only approved features; document transformations; create missingness indicators only when justified; save the feature table with a data dictionary.

End-of-day product: Feature table v1 and feature dictionary.

Quality gate: Each feature has a source, unit or encoding, transformation rule, and leakage status.

Day 37: Chemical Traits as Optional ML Features

Time design: Full day: 30 min check-in; 90 min trait-to-feature caution; 120 min feature integration lab; 120 min evidence audit; 45 min log.

Purpose: Use uafR-derived features only when the chemical evidence and sample mapping are defensible.

Tools and materials: Saved uafR trait matrix if available, Trait Evidence Audit, feature table, and chemical claim-limit memo.

Work blocks: Decide whether chemical traits enter the model; if they do, join by documented sample or compound mapping and preserve missingness; if not, explain why.

End-of-day product: Chemical feature decision note.

Quality gate: No compound trait enters the model without a traceable link to sample-level evidence and source-backed trait evidence.

Day 38: Advanced Validation After the Beginner Model

Time design: Full day: 30 min objective; 90 min model comparison concepts; 120 min advanced validation run; 120 min diagnostics; 60 min writing.

Purpose: Compare the beginner model against stricter validation checks without losing interpretability.

Tools and materials: Beginner ML outputs, `scripts/05_ml_01_advanced_validation.R`, `ml_resources/advanced_ml_extension_guide.md`, model-card template, and metrics table.

Work blocks: Confirm that the beginner ladder is understood; run the advanced validation script; compare time-holdout and leave-one-site-out results; record where blocked validation weakens or supports the beginner result.

End-of-day product: Advanced validation folds, split comparison table, and validation notes.

Quality gate: All comparisons use documented targets, eligible rows, and feature sets; any skipped model is treated as evidence of overfitting risk rather than a failure to hide.

Day 39: Spatial and Temporal Context

Time design: Full day: 30 min check-in; 90 min spatial/temporal concepts; 120 min site/date summaries; 120 min model-risk review; 45 min log.

Purpose: Understand how site, season, tide, rainfall, and repeated sampling can structure model results.

Tools and materials: Site/date metadata, feature table, map or site-context notes, and leakage audit.

Work blocks: Summarize target and predictors by site/date/context; identify whether site identifiers act as shortcuts; decide whether site/date features belong in the main model or sensitivity analysis.

End-of-day product: Spatial/temporal context memo.

Quality gate: The memo distinguishes environmental context from leakage-prone identifiers.

Day 40: Sensitivity, Permutation Importance, and Leakage Review

Time design: Full day: 30 min objective; 90 min leakage audit; 90 min guided sensitivity run; 120 min output review; 60 min sign-off.

Purpose: Test whether feature-set changes improve performance for defensible reasons or because they leak information.

Tools and materials: Feature table, split plan, advanced sensitivity script, leakage checklist, and research log.

Work blocks: Run the sensitivity script; compare the core feature set with conductivity and context sensitivity outputs; inspect permutation importance; write why a high-importance feature is not causal proof.

End-of-day product: Feature-set sensitivity table, permutation-importance table, sensitivity warnings, and leakage-audit memo.

Quality gate: Conductivity or context features are not promoted to the main model unless the project question justifies their availability and interpretation.

Day 41: Weekend Matrix Sparsity and Missing-Data Challenge

Time design: Weekend: 1 to 3 hours.

Purpose: Evaluate whether sparse features are helping or just adding noise.

Tools and materials: Feature table, missingness report, optional trait matrix, and model card.

Work blocks: Calculate missingness or prevalence for each feature; identify sparse features; propose exclusions or sensitivity checks.

End-of-day product: Sparsity and missingness sensitivity note.

Quality gate: Any proposed feature removal is justified before seeing whether it improves the preferred result.

Day 42: Weekend Advanced Model Review

Time design: Weekend: 1 to 2.5 hours.

Purpose: Translate Week 6 results into cautious interpretation.

Tools and materials: Advanced model outputs, `scripts/05_ml_03_advanced_model_report.R`, decision-gate table, leakage audit, and research log.

Work blocks: Run the advanced report script; open the model review and decision gate; write a one-page memo naming the most defensible model, why it was chosen, and what it cannot support.

End-of-day product: Advanced model review, decision gate, and model interpretation memo.

Quality gate: The memo discusses uncertainty, leakage, split design, and limitations at least as clearly as performance.

3.7 Week 7: Visualization, Evidence Synthesis, and Communication**Week 7: Visualization and evidence synthesis**

Project milestone: Students turn processed data and model outputs into figures, tables, and defensible scientific interpretation.

Primary product: Figure package, claim-evidence-limitation memo, literature synthesis table, and draft report sections.

Main tool emphasis: Visual honesty, missingness, model diagnostics, source evidence, literature integration, and reviewer-style critique.

Day 43: Figure Planning Before Plotting

Time design: Full day: 30 min check-in; 90 min visual-purpose clinic; 120 min figure planning; 120 min independent sketching; 45 min review.

Purpose: Plan figures around claims, not around available plotting options.

Tools and materials: Figure Planning Worksheet, final feature table, model outputs, and source evidence.

Work blocks: Define the question for each figure; identify audience, variables, grouping, missingness, caption claim, and caveat.

End-of-day product: Figure plan for at least four candidate figures.

Quality gate: Each planned figure has a purpose, data source, caption draft, and claim limit.

Day 44: Water-Quality and Source-Coverage Figures

Time design: Full day: 30 min objective; 90 min plotting principles; 120 min guided plotting; 120 min project plotting; 60 min log.

Purpose: Create figures that show water-quality patterns and data coverage honestly.

Tools and materials: Cleaned data, missingness report, source coverage table if available, and plotting script.

Work blocks: Build summaries by site/date/context; plot salinity/conductivity/DO/nutrients with units; create a missingness or source-coverage heatmap.

End-of-day product: Two draft water-quality/source-coverage figures.

Quality gate: Axes, units, sample counts, missingness, and grouping definitions are visible.

Day 45: Model Diagnostic and Interpretation Figures

Time design: Full day: 30 min check-in; 90 min diagnostics; 120 min model plot lab; 120 min independent revision; 45 min checkpoint.

Purpose: Visualize model performance without exaggerating accuracy.

Tools and materials: Model metrics, predictions, model card, and figure caption checklist.

Work blocks: Plot observed versus predicted values, residuals, confusion matrix, or probability distributions as appropriate; write captions that state split design and limits.

End-of-day product: Two draft model figures and caption drafts.

Quality gate: No figure reports training-only performance as final evidence.

Day 46: Figure Clinic and Peer Review

Time design: Full day: 30 min objective; 90 min peer critique; 90 min revision; 120 min caption repair; 60 min checkpoint.

Purpose: Improve figures through reviewer-style feedback.

Tools and materials: Figure Caption Checklist, draft figures, peer review form, and claim memo.

Work blocks: Review axes, legends, color choices, caption claims, missingness, and accessibility; revise figures and captions.

End-of-day product: Revised figure package v1.

Quality gate: A reviewer can identify the data, variables, sample counts, and claim limit from the figure and caption.

Day 47: Literature-to-Evidence Bridge

Time design: Full day: 30 min check-in; 90 min literature strategy; 120 min source extraction; 120 min writing; 45 min review.

Purpose: Connect project results to literature without overstating similarity.

Tools and materials: Literature and Evidence Synthesis worksheet, annotated readings, project figures, and model card.

Work blocks: Search by variables, site context, organism/system, method, and model target; extract system, method, chemical/water evidence, main claim, and limitation.

End-of-day product: Literature synthesis table and one claim-evidence-limitation paragraph.

Quality gate: The paragraph cites evidence and states how the student's dataset is similar or different.

Day 48: Weekend Figure Revision and Export

Time design: Weekend: 1 to 3 hours.

Purpose: Prepare figures for final report drafting.

Tools and materials: Draft figures, captions, README, and figure export settings.

Work blocks: Export figures with consistent names; update captions; record script and output paths in README.

End-of-day product: Figure package v2.

Quality gate: Each figure can be regenerated and has a matching caption.

Day 49: Weekend Visual Narrative Outline

Time design: Weekend: 1 to 2 hours.

Purpose: Turn figures into the skeleton of the final report.

Tools and materials: Figure package, literature synthesis table, model interpretation memo.

Work blocks: Order figures; write one sentence per figure explaining what it contributes to the story.

End-of-day product: Visual narrative outline.

Quality gate: The outline does not contain unsupported causal or confirmation claims.

3.8 Week 8: Capstone Report, Presentation, and Handoff

Week 8: Final research product and handoff

Project milestone: Students finish a reproducible research package that another researcher can inspect and continue.

Primary product: Manuscript-style report, presentation/poster, reproducible folder, final package audit, continuation note, and exit reflection.

Main tool emphasis: Scientific writing, reproducibility audit, presentation, peer review, archive discipline, and next-step planning.

Day 50: Report Storyboard

Time design: Full day: 30 min objective; 90 min report structure; 120 min storyboard lab; 120 min drafting; 45 min checkpoint.

Purpose: Organize the final report before writing full prose.

Tools and materials: Report template, visual narrative outline, project brief, figures, and model card.

Work blocks: Draft title, abstract bullets, introduction rationale, methods sections, results sequence, discussion claims, limitations, and reproducibility statement.

End-of-day product: Report storyboard.

Quality gate: Every planned result connects to a figure/table and every claim connects to evidence.

Day 51: Methods From Scripts and Logs

Time design: Full day: 30 min check-in; 90 min methods standards; 120 min script-to-methods lab; 120 min writing; 60 min review.

Purpose: Write methods that are specific enough to rerun.

Tools and materials: README, scripts, data dictionary, QA/QC log, uafR query settings, model card, and session info.

Work blocks: Write data sources, cleaning rules, uafR settings if used, feature engineering, train/test split, model type, metrics, and software versions.

End-of-day product: Methods section draft.

Quality gate: A reader can match each methods paragraph to a script, file, or documented setting.

Day 52: Results Without Overclaiming

Time design: Full day: 30 min objective; 90 min results-writing clinic; 120 min paragraph drafting; 120 min table/figure checks; 45 min log.

Purpose: Write results that describe observations and model outputs without causal inflation.

Tools and materials: Figures, tables, model metrics, claim-evidence-limitation memo, and caption checklist.

Work blocks: Write results paragraphs tied to figure numbers; include sample sizes, units, model split, and uncertainty; remove unsupported explanations.

End-of-day product: Results section draft.

Quality gate: Each result paragraph describes what was observed or predicted and avoids unsupported mechanism claims.

Day 53: Discussion, Limitations, and Next Steps

Time design: Full day: 30 min check-in; 90 min discussion clinic; 120 min writing; 90 min peer critique; 60 min revision.

Purpose: Explain why the results matter while keeping uncertainty visible.

Tools and materials: Literature synthesis table, model interpretation memo, chemical claim-limit memo, and project risk note.

Work blocks: Write discussion paragraphs that connect to WiCCED context; include limitations for data, chemistry, model validation, and generalization; define follow-up tests.

End-of-day product: Discussion and limitations draft.

Quality gate: Limitations are concrete and tied to the actual dataset and methods.

Day 54: Final Reproducibility Audit

Time design: Full day: 30 min objective; 90 min audit protocol; 150 min clean-session rerun; 90 min fixes; 45 min sign-off.

Purpose: Verify the project can be rerun from a clean session.

Tools and materials: `scripts/run_wicced_smoke_tests.R`, project scripts, README, final package audit worksheet, and session info.

Work blocks: Restart R; rerun scripts; compare expected outputs; fix path or missing-file problems; document remaining risks.

End-of-day product: Final reproducibility audit and session information.

Quality gate: All required outputs exist or a clear not-applicable note explains why.

Day 55: Presentation or Poster Build

Time design: Full day: 30 min check-in; 90 min presentation structure; 120 min slide/poster build; 90 min practice; 60 min revision.

Purpose: Create a clear oral or poster presentation around one central result.

Tools and materials: Report draft, figure package, model card, and presentation planning worksheet.

Work blocks: Build slides/poster with question, data, methods, main result, model validation, limitations, and next steps; practice a 5 to 8 minute version.

End-of-day product: Presentation/poster draft and speaking notes.

Quality gate: The presentation makes one central argument and includes limitations without hiding them.

Day 56: Final Presentation, Archive, and Continuation Note

Time design: Full day: 30 min setup; 90 min presentations; 90 min Q&A and feedback; 120 min final archive; 60 min exit reflection.

Purpose: Complete the program with a usable research package and a clear path for continuation.

Tools and materials: Final report, presentation, project folder, final package audit, continuation note, and research log.

Work blocks: Present the project; answer reviewer-style questions; archive the package; write unresolved issues and recommended next experiments.

End-of-day product: Final report, presentation, archive, continuation note, and exit reflection.

Quality gate: A future student can open the folder, understand the project, rerun the scripts, and see exactly what remains unresolved.

4 uafR Workflows for Water-Quality Research

uafR belongs in this water-quality program when chemical screening evidence is available or when a chemical query list needs structured public-database context. The package is especially useful for GC-MS workflows and source-backed chemical trait extraction. It should be used cautiously and documented as one evidence layer, not as the entire water-quality analysis.

4.1 Verified Function Surface

The following function names and output families were verified from the repository source, documentation, tests, or existing training scripts before inclusion:

- `spreadOut()` for reshaping and summarizing GC-MS hit-table output.
- `mzExacto()` for extracting selected chemicals from `spreadOut()` output.
- `exactoThese()` for building query lists from `categorate()` output.
- `categorate(detail = "research")` for research-grade chemical enrichment.
- `validateCategorateResult()` for result validation.
- `chemicalTraitMatrix()` for source-backed trait matrices.
- `chemicalTraitEvidence()` for evidence trails behind traits and matrix keys.
- `chemicalTraitOntology()` and `chemicalTraitOntologyMatrix()` for controlled trait terms when available.
- `pubchemProfile()` for targeted PubChem identity/property profiles.

4.2 Workflow Decision Tree

When to Use uafR

Use uafR if the project has GC-MS hit tables, a defensible list of candidate chemicals, or a need to turn chemical names into source-backed descriptors. Do not use uafR as a substitute for salinity, dissolved oxygen, nutrient, bacterial, or field-sensor measurements.

Project situation	Recommended uafR action	Required caution
GC-MS hit table exists and column names match expected format	Use <code>spreadOut()</code> followed by <code>mzExacto()</code> for selected compounds.	Treat matches as tentative unless confirmed by standards and method evidence.
Chemical names exist but no sample-level GC-MS evidence exists	Use <code>pubchemProfile()</code> or a small <code>categorate()</code> query for source context only.	Do not imply sample presence.
Many compounds need grouping for exploratory comparison	Use <code>categorate(detail = "research")</code> and <code>chemicalTraitMatrix()</code> after validation.	Use cached, throttled, small batches and document source coverage.
Traits are needed for model features	Use <code>chemicalTraitEvidence()</code> and a grouping dictionary before modeling.	Do not include narrative fields or unsupported traits as features.
No chemical data or chemical question exists	Write a not-applicable note.	Do not force chemical analysis into a water-quality project.

4.3 Simulated GC-MS Teaching Workflow

The manual includes a simulated WiCCED-themed GC-MS hit table in the `data/` folder. It is not a real environmental dataset. It is designed to teach the mechanics and interpretation value of uafR-style GC-MS processing while staying scientifically honest.

The simulated table contains the required uafR columns: `Component.RT`, `Component.Area`, `Base.Peak.MZ`, `File.Name`, `Compound.Name`, and `Match.Factor`. It also includes sample-context columns so students can connect the chemical patterns to water-quality scenarios. The pattern includes taste/odor candidates, agricultural screening candidates, urban/stormwater markers, marsh/aquaculture organic-matter signals, plant-associated volatile background, and duplicate detections that require aggregation.

From the `training_wicced_water_quality/` folder, run:

```
Rscript scripts/03_uafR_chemical_screening_template.R
```

The default mode is offline and deterministic. It validates the uafR-compatible columns, creates a spread object with the same major structure used downstream by uafR, aggregates exact-name matches for the simulated query compounds, computes relative abundance within each sample, and writes summary files.

Expected outputs in `results/` are:

- `wicced_simulated_spread.rds`
- `wicced_simulated_exact_matches.csv`
- `wicced_simulated_long_abundance.csv`
- `wicced_simulated_sample_group_summary.csv`
- `wicced_simulated_compound_evidence_notes.csv`
- `wicced_simulated_gcms_pipeline_summary.txt`

Students should inspect whether duplicate **Caffeine** detections in the urban stormwater sample were aggregated, whether relative abundance sums to 1 within each sample, and whether chemical patterns differ by sample context. The correct conclusion is not that those compounds occur in Delaware water. The correct conclusion is that the pipeline can convert a GC-MS hit table into auditable, sample-level chemical evidence that could guide real follow-up work.

Live uafR Mode

The script also accepts a final `live` argument that calls real `spreadOut()` and `mzExacto()` when `uafR` is installed or loadable and web services are available. Live mode may query PubChem/NCI and should be used only as a small smoke test with clear documentation. The offline simulated mode is the required training path.

4.4 Research Enrichment Template

Live enrichment can be slow, dependency-sensitive, and affected by public service availability. The safe pattern is a tiny smoke test, cached output, and validation before scaling.

```
library(uafR)

data("library_data", package = "uafR")

compounds <- c("caffeine", "atrazine")

result <- categorate(
  compounds = compounds,
  chemical_library = library_data,
  detail = "research",
  cache = TRUE,
  throttle = 0.2,
  trait_matrix_profile = "core",
  trait_matrix_min_confidence = "medium",
  trait_matrix_max_traits = 100
)

validation <- validateCategorateResult(result)
names(result)
result$ValidationSummary
result$SourceCoverage
head(result$ChemicalTraits)

saveRDS(result, file = "results/categorate_research_small.rds")
```

This code is verified as a package-level pattern, but it should be treated as a live-query template. If a machine lacks network access or required chemical back-end packages, students should inspect a saved result object instead of forcing a live run.

4.5 Trait Evidence Before Analysis

Trait matrices are useful only when the student understands where the columns came from. Before any trait enters a figure, clustering, or model, the evidence should be traceable.

```
traits <- chemicalTraitMatrix(  
  result,  
  profile = "core",  
  mode = "binary",  
  min_confidence = "medium",  
  max_traits = 100  
)  
  
evidence <- chemicalTraitEvidence(result, min_confidence = "medium")  
  
head(traits)  
head(evidence)  
write.csv(traits, "results/chemical_trait_matrix.csv", row.names = FALSE)  
write.csv(evidence, "results/chemical_trait_evidence.csv", row.names = FALSE)
```

The trait evidence table should be used to fill the Trait Grouping Dictionary. The student should record source table, source field, evidence text or identifier, extraction rule, confidence, allowed values, and caveat.

4.6 Water-Quality Integration Pattern

If chemical traits are used with water-quality measurements, the join must be explicit. The student must be able to answer:

- What sample, bottle, extract, or file links the chemical data to the water-quality row?
- Were chemical features measured from the same sample, site, date, or treatment?
- Are abundance values raw peak areas, relative areas, standardized values, or presence/absence indicators?
- Are unresolved compounds retained as missing, excluded, or summarized separately?
- Could the chemical feature reveal the target variable in a way that creates leakage?

If these questions cannot be answered, chemical traits should remain an evidence appendix rather than model features.

4.7 Required Documentation

Every uafR-including project should include:

- Input file inventory.
- Query settings log.
- Saved raw result object.
- Validation output.

- Source coverage report.
- Evidence table for retained traits.
- Not-applicable notes for excluded tables or unsupported sources.
- A claim-limit memo that states what cannot be concluded.

5 Machine-Learning Guidance for Water Quality

Machine learning in this program is a research tool, not a shortcut around scientific judgment. Models are useful when they help evaluate patterns in water-quality measurements, prioritize follow-up questions, or compare feature sets. Models are not acceptable when they obscure data problems, inflate weak evidence, or produce claims that the sampling design cannot support.

5.1 Beginner Script Ladder

Students should start with the four-script ladder before modifying any model. Each script is intentionally small enough to inspect, and each one writes outputs that can be opened in RStudio.

Run these commands from the `training_wicced_water_quality/` folder:

```
Rscript scripts/04_ml_01_build_feature_table.R
Rscript scripts/04_ml_02_explore_features.R
Rscript scripts/04_ml_03_train_models.R
Rscript scripts/04_ml_04_diagnostics_and_model_card.R
```

Step	What it teaches	Key outputs
Build feature table	Targets, features, exclusions, leakage notes, one-hot context variables, and time-based train/test split.	Feature table, feature dictionary, split plan, training rows, testing rows.
Explore features	Summary statistics, class balance, simple correlations, and plots before modeling.	Feature summary, correlation table, class balance, exploration plots, student questions.
Train models	Mean-only regression baseline, linear regression, majority-class baseline, logistic classifier, metrics, predictions, and coefficients.	Regression metrics, classification metrics, predictions, coefficients, model-comparison summary.
Diagnose and document	Prediction plots, residuals, confusion-matrix thinking, model-card writing, and interpretation limits.	Diagnostic plots, diagnostic summary, model card, interpretation guide.

Beginner Rule

The first ML goal is not to build the most accurate model. The first goal is to build a model that can be inspected, compared to a baseline, explained without overclaiming, and rerun by another student.

5.2 Guiding Resources

The `ml_resources/` folder contains beginner support files:

- `README.md`: the beginner path and required interpretation rule.
- `ml_glossary.csv`: plain-language definitions for target, feature, leakage, baseline, residual, model card, and common metrics.
- `ml_workflow_checklist.csv`: a checklist from question to handoff.
- `ml_troubleshooting.md`: common beginner problems and how to reason through them.
- `model_card_field_guide.md`: what to write in each model-card section.

5.3 What Students Should Open First

After the four scripts run, students should open these files in order:

1. `results/ml/ml_feature_dictionary.csv`. Confirm which variables are targets, predictors, excluded variables, and metadata.
2. `results/ml/ml_train_test_split_plan.csv`. Explain what the split tests.
3. `results/ml/ml_exploration_questions.md`. Answer each question in the research log.
4. `results/ml/ml_regression_metrics.csv`. Compare the linear model to the mean-only baseline.
5. `results/ml/ml_classification_metrics.csv`. Compare the logistic model to the majority-class baseline.
6. `results/ml/ml_predictions.csv`. Find rows where the model missed.
7. `results/ml/ml_model_card.md`. Revise the model card for the student's project.

5.4 Beginner Feature Policy

The teaching scripts deliberately exclude `conductivity_ms_cm` from the beginner salinity and high-salinity models because conductivity is closely related to salinity. That exclusion teaches leakage caution. Conductivity can be added later only as a documented sensitivity model when the research question says it is available and scientifically appropriate.

5.5 Advanced Extension After Beginner Mastery

The advanced extension is required only after the beginner ladder has been fully understood. It should not be used as a shortcut to make a model look more impressive. Before starting the advanced scripts, students must be able to explain the target, the included features, excluded variables, leakage risks, train/test split, baseline comparison, prediction errors, and the beginner model card.

Run these commands from the `training_wicced_water_quality/` folder only after the four beginner ML scripts have been completed:

```
Rscript scripts/05_ml_01_advanced_validation.R
Rscript scripts/05_ml_02_permutation_importance_and_sensitivity.R
Rscript scripts/05_ml_03_advanced_model_report.R
```

Advanced step	What it adds	Key outputs
Blocked validation	Compares the beginner feature set and sensitivity feature sets under a time holdout and leave-one-site-out folds.	Validation folds, split comparison, validation notes.
Sensitivity and importance	Tests conductivity as a flagged sensitivity feature, skips overfit feature sets, and estimates held-out permutation importance.	Feature-set metrics, permutation importance, sensitivity warnings, sensitivity plots.
Advanced review	Converts advanced outputs into a decision gate and written interpretation.	Advanced model review, decision gate, final recommendations.

Advanced ML Gate

An advanced model is not better because it is more complex. It is better only if it improves held-out performance, survives leakage review, remains interpretable, and matches the scientific question. A sensitivity result that improves performance can still be inappropriate for the main analysis.

The advanced extension writes outputs to the advanced ML results folder. Students should open the model-review markdown file first, then the decision-gate CSV, then the validation and permutation-importance tables. Any gate marked as requiring review must be resolved before advanced ML outputs appear in a final report.

The `ml_resources/advanced_ml_extension_guide.md` file gives the student-facing sequence and interpretation rules for this extension.

5.6 Acceptable ML Question Types

Question type	Example	Good first model
---------------	---------	------------------

Regression	Predict salinity, dissolved oxygen, nitrate, or chlorophyll from available environmental features.	Mean-only baseline, linear model, regularized model, or tree model with careful validation.
Classification	Classify samples as high-salinity versus lower-salinity using a pre-defined threshold.	Majority-class baseline, logistic regression, decision tree, or random forest if data volume supports it.
Anomaly screening	Flag observations that are unusual relative to the student's dataset.	Rule-based flags first; simple distance or residual screening only after QA/QC.
Feature-set comparison	Compare water-quality variables alone versus water-quality plus source-backed chemical traits.	Same split and same target across feature sets.
Exploratory grouping	Cluster samples or compounds to identify patterns for follow-up.	PCA or clustering as exploratory analysis with no causal claim.

5.7 Leakage Control

Leakage occurs when information reaches the model that would not be available at prediction time or that directly encodes the target. Leakage makes performance look better than it is.

Common water-quality leakage examples:

- Predicting salinity while including conductivity if the purpose is to predict salinity without a salinity-related sensor. Conductivity may be valid in some designs, but the relationship must be explicit.
- Including site ID when the train and test sets contain the same sites and the target is mostly site-specific.
- Randomly splitting repeated measurements from the same site/date when the goal is generalization to new dates or sites.
- Including post-treatment, post-event, or manually interpreted variables that would not be available during deployment.
- Including chemical traits that were selected after inspecting the target response.

5.8 Split Designs

Split design	Use when	Caution
Random split	Rows are independent and the question is narrow.	Often too optimistic for repeated site/date data.
Time-based split	The project asks whether earlier data predict later observations.	Seasonal shifts can dominate performance.
Site-holdout split	The project asks whether a model generalizes to new sites.	Requires enough sites and careful interpretation.
Group split	Samples are nested by site, date, batch, instrument run, or treatment.	Groups must be defined before modeling.

Cross-validation	Data are limited but model comparison is needed.	Folds must respect groups if repeated measures exist.
------------------	--	---

5.9 Metrics

Regression models should report at least one scale-dependent metric and one visual diagnostic:

- RMSE: penalizes larger errors.
- MAE: average absolute error in the target unit.
- Test-set observed-versus-predicted plot.
- Residual plot, especially across site/date or predicted value.

Classification models should report:

- Confusion matrix.
- Accuracy only with class balance reported.
- Sensitivity and specificity when false negatives and false positives matter.
- Threshold used to convert probabilities into classes.
- If probabilities are used, a probability distribution or calibration check.

5.10 Feature Engineering Rules

Feature engineering should be boring, documented, and defensible. Good features are created from measurements or source-backed records that would be available at prediction time. Transformations should be simple unless there is a strong reason.

Rules:

- Freeze the analysis row set before comparing models.
- Keep raw variables recoverable.
- Record units before scaling or transformation.
- Avoid transformations that depend on the whole dataset if the model would later be deployed.
- Treat missingness as information only when the missingness mechanism is plausible and documented.
- Use source-backed chemical traits only after an evidence audit.

5.11 Model Cards

Every model must have a model card. A model card is a short technical record that states:

- Project question.
- Prediction target.

- Rows included and excluded.
- Features included.
- Train/test or cross-validation design.
- Baseline model.
- Candidate model.
- Metrics.
- Diagnostic plots.
- Known limitations.
- Use restrictions.
- Recommended next validation step.

5.12 Interpretation Language

Good model wording:

- “Under the recorded time-based split, the baseline model predicted salinity with lower test-set MAE than the mean-only baseline.”
- “The feature importance summary is treated as a hypothesis-generating screen because correlated predictors and site structure can distort apparent importance.”
- “The classification threshold favors sensitivity, which increases false positives and is appropriate only for screening.”

Weak model wording:

- “The model proves which variable causes poor water quality.”
- “High accuracy means the model is ready for field deployment.”
- “This feature is important, so it must be the biological mechanism.”

6 Lab Compendium

Each lab is designed to produce a reusable project artifact. Labs can be run with real approved data, the teaching data included with this manual, or a saved uafR example. When real data are not ready, the teaching-data path should be used rather than delaying the skill.

6.1 Lab 1: WiCCED Chemical and Water-Quality Question Triage

Objective

Turn a broad interest into a feasible 8-week water-quality research question.

Inputs: WiCCED context notes, Project Brief worksheet, evidence map, and research log.

Procedure: Write three broad interests. For each, name the sample unit, response variable, candidate predictors, data source, and likely evidence limit. Reject questions that require unavailable data, direct causation, or confirmed chemical identity without standards. Select one minimum viable project and one advanced extension.

Expected output: Approved Project Brief v1.

Quality gate: The question can be answered with available or teaching data in eight weeks and has a clear claim boundary.

Troubleshooting: If the question is too broad, reduce the system, time range, variable count, or claim strength. If the data are unavailable, switch to teaching data and write a continuation note for the real data.

6.2 Lab 2: Reproducible Project Build

Objective: Build a project folder that can be handed to another student.

Inputs: `scripts/00_setup_check.R`, `scripts/01_project_setup.R`, RStudio, and chosen project root.

Procedure: Run the setup check. Run the project setup script in a selected folder. Inspect the created directories. Add a README section for project question, data rules, script order, and current status. Restart R and rerun the setup check.

Expected output: Project skeleton, README, setup log, and session note.

Quality gate: No step depends on unstated console objects or hidden local paths.

6.3 Lab 3: Water-Quality Data Dictionary

Objective: Define every column before analysis.

Inputs: Teaching or project data, Variable Dictionary worksheet, and QA/QC notes.

Procedure: Record column name, plain-language definition, unit, type, source, method, missing-value code, expected range, and interpretation caveat. Mark target variables and predictors separately.

Expected output: Water-quality data dictionary v1.

Quality gate: Every model feature can be traced to a dictionary row.

6.4 Lab 4: Field and Lab QA/QC Audit

Objective: Flag suspicious water-quality records without destroying raw data.

Inputs: `scripts/02_water_quality_qaqc.R`, data dictionary, QA/QC Flag Log worksheet.

Procedure: Run the QA/QC script on the teaching data. Review range checks for salinity, conductivity, dissolved oxygen, pH, turbidity, nitrate, chlorophyll, and temperature. Adapt rules only after documenting why. Save a flagged copy and summary.

Expected output: QA/QC flags CSV and summary CSV.

Quality gate: Raw values remain recoverable and every flag has a rule.

6.5 Lab 5: Database Source Audit

Objective: Learn what each source contributes before using database fields.

Inputs: Database Evidence Map worksheet, uafR documentation, and one selected chemical or water-quality source.

Procedure: For each source, record source type, query term, returned identifiers or fields, what it can support, what it cannot support, and how it could enter analysis.

Expected output: Database Evidence Map v1.

Quality gate: No source is listed only as “useful”; each has a specific evidence role.

6.6 Lab 6: Simulated Core uafR-Compatible Workflow

Objective: Practice GC-MS hit-table validation, spreading, exact-match aggregation, and abundance interpretation without live web services.

Inputs: the simulated GC-MS hit table, simulated query list, simulated compound reference, simulated sample metadata, `scripts/03_uafR_chemical_screening_template.R`, and research log. All four simulated data files live in the `data/` folder.

Procedure: Run the script from the manual folder. Inspect the raw hit table and verify the six required uafR columns. Open the exact-match output and confirm that duplicate urban-stormwater **Caffeine** detections are aggregated. Open the long-abundance output and confirm that relative abundance sums

to 1 within each sample. Compare the sample-group summary across freshwater, agricultural, urban, transition, marsh, and aquaculture contexts.

Expected output: Simulated spread RDS, exact-match abundance table, long relative-abundance table, group summary, compound evidence notes, and workflow summary.

Quality gate: The student can explain the input columns, the aggregation step, the relative-abundance calculation, the context patterns, and why the simulated results are not evidence about real Delaware waters.

6.7 Lab 7: Research Enrichment Smoke Test

Objective: Safely plan or run a tiny `categorate(detail = "research")` enrichment.

Inputs: Query Settings Log, `library_data`, saved result if available, and cache/throttle settings.

Procedure: Choose one to three compounds. Record query spelling, cache directory, throttle, profile, and purpose. If live services are available, run a small query. If not, inspect a saved result object. Validate the result before interpretation.

Expected output: Query settings log, validation summary, and source coverage note.

Quality gate: The student inspects validation outputs before any traits are used.

6.8 Lab 8: Evidence Court

Objective: Trace claims back to source evidence.

Inputs: `uafR` result or saved example, `chemicalTraitEvidence()`, Evidence Court worksheet.

Procedure: Select five candidate traits or claims. For each, record source table, source field, evidence text or identifier, confidence, allowed interpretation, and forbidden interpretation.

Expected output: Evidence Court table.

Quality gate: Every retained claim has source-backed evidence and a caveat.

6.9 Lab 9: ML Feature Table Build

Objective: Build the modeling table from approved variables only.

Inputs: Cleaned data, feature inventory, optional chemical trait matrix, train/test split plan, `scripts/04_ml_01_build_feature_table.R`, and `ml_resources/ml_workflow_checklist.csv`.

Procedure: Run the feature-table script on the teaching data. Open the feature table, feature dictionary, split plan, training rows, and testing rows. Confirm that targets are not used as predictors. Confirm that conductivity is excluded from the beginner salinity model as a leakage-caution exercise. For project data, modify only after the teaching outputs are understood.

Expected output: Feature table, feature dictionary, split plan, training rows, and testing rows.

Quality gate: No feature is included without source, transformation rule, and leakage status.

6.10 Lab 10: Beginner ML Ladder

Objective: Fit defensible first models, compare them to simple baselines, and create a model card.

Inputs: the ML exploration script, model-training script, diagnostics/model-card script, feature table, split plan, glossary, troubleshooting guide, and model-card field guide.

Procedure: Run the exploration script and answer the exploration questions. Run the training script and compare regression to a mean-only baseline and classification to a majority-class baseline. Run the diagnostics script and inspect prediction plots, residuals, class predictions, coefficients, diagnostic summary, and model card.

Expected output: Exploration plots, feature summaries, metrics CSVs, predictions CSV, coefficients CSV, diagnostic plots, diagnostic summary, interpretation guide, and model-card draft.

Quality gate: The model is evaluated on held-out data and limitations are recorded.

6.11 Lab 11: Leakage and Validation Drill

Objective: Detect invalid model evaluation before final interpretation and run the advanced ML extension only after the beginner ladder is understood.

Inputs: Feature table, split plan, model-card draft, leakage checklist, advanced ML guide, blocked-validation script, sensitivity/permutation script, and advanced report script.

Procedure: First, confirm that the beginner outputs can be explained without reading from the screen. Then search for duplicate rows, repeated site/date structure, post-outcome variables, target-derived predictors, and features chosen after inspecting the outcome. Run the advanced validation script and compare time-holdout results to leave-one-site-out results. Run the sensitivity script and inspect whether conductivity or context features change performance. Run the advanced report script and open the decision gate before changing the preferred model.

Expected output: Leakage audit memo, advanced validation folds, split-comparison table, feature-set sensitivity table, permutation-importance table, advanced model review, and decision gate.

Quality gate: The final split tests the kind of generalization the project claims, and any conductivity or context improvement is treated as a review item rather than automatic model approval.

6.12 Lab 12: Source Coverage Missingness Audit

Objective: Show where evidence is strong, weak, or absent.

Inputs: Missingness report, uafR source coverage if available, water-quality data dictionary.

Procedure: Calculate missingness by variable and group. For chemical evidence, summarize source coverage by compound. Build a heatmap or summary table.

Expected output: Source coverage and missingness summary.

Quality gate: Missing data are discussed before model or figure claims.

6.13 Lab 13: Matrix Sparsity Sensitivity Analysis

Objective: Test whether sparse features are useful or misleading.

Inputs: Feature table, optional trait matrix, beginner model outputs, advanced sensitivity outputs, and model card.

Procedure: Count feature prevalence and missingness. Define a pre-model exclusion threshold. Use the advanced sensitivity outputs to decide whether a sparse or leakage-prone feature changes results for a defensible reason. Rerun one model with sparse features removed only after recording the rule.

Expected output: Sensitivity note and revised model card.

Quality gate: Exclusions are justified before comparing performance.

6.14 Lab 14: Figure Clinic

Objective: Improve figures so they communicate evidence honestly.

Inputs: Draft figures, caption checklist, peer review form, and source evidence.

Procedure: Review each figure for purpose, axes, units, sample counts, missingness, grouping, color, caption claim, and caveat. Revise at least two figures.

Expected output: Revised figure package.

Quality gate: Captions state what the figure supports and what it does not.

6.15 Lab 15: Reviewer Response Exercise

Objective: Practice responding to scientific critique.

Inputs: Draft report, peer review form, and model card.

Procedure: Receive three reviewer-style comments: one about data, one about model validity, and one about claim strength. Write a response and revise the report or note why revision is not appropriate.

Expected output: Reviewer response table.

Quality gate: Responses point to concrete edits, files, or evidence.

6.16 Lab 16: Final Package Recovery Exercise

Objective: Confirm that the final project can survive handoff.

Inputs: Final project folder, README, scripts, data, results, figures, and final audit worksheet.

Procedure: Start from a clean R session. Follow README steps. Confirm outputs exist. Record missing files, broken paths, or unclear instructions. Fix what can be fixed and document what remains.

Expected output: Final package audit and continuation note.

Quality gate: A future student can rerun the project or understand exactly why a step cannot be rerun.

7 Worksheets

Worksheets are intended as student artifacts. They may be printed, copied into a lab notebook, or reproduced as CSV/Markdown files. Every worksheet should include the date, student name, project folder, and current file path.

7.1 Project Brief

Purpose

Define a feasible water-quality research question and the minimum viable project.

Use during: Week 1, then revise at every decision gate.

Required fields: project title; student name; co-advisers; research question; sample unit; target variable; candidate predictors; project data source; teaching-data fallback; optional uafR role; ML target; expected figures; expected final claim; known limitations; minimum viable product; advanced extension.

Common mistakes: asking a causal question without a causal design; choosing a target that is not measured; including too many systems; treating tentative chemical matches as confirmed.

Completion standard: A reviewer can tell what will be analyzed, why it matters to WiCCED, what data are needed, and what claim will remain out of bounds.

7.2 Water-Quality Variable Dictionary

Purpose: Make every variable interpretable before analysis.

Required fields: column name; plain-language name; unit; variable type; source file; method; sample unit; expected range; missing-value code; detection limit if relevant; target/predictor/excluded status; caveat.

Completion standard: Every feature used in a model appears in the dictionary.

7.3 Site and Sample Metadata Sheet

Purpose: Preserve context that affects interpretation.

Required fields: site ID; site name; latitude/longitude or protected location code; date/time; depth; weather/rainfall note; tidal or hydrologic context if available; sampler; instrument; method; project

restrictions; data privacy status.

Completion standard: The sample unit can be reconstructed without relying on memory.

7.4 QA/QC Flag Log

Purpose: Record suspicious values without deleting raw observations.

Required fields: file; row ID; variable; raw value; flag type; rule; reason; action taken; replacement value if any; reviewer; date.

Completion standard: Every changed or excluded value can be traced back to the raw value and decision rule.

7.5 Database Evidence Map

Purpose: Clarify what each source can support.

Required fields: source name; query term; source type; returned field; identifier; what it supports; what it cannot support; citation or URL; planned use in analysis.

Completion standard: No source is used in a claim without a recorded limitation.

7.6 GC-MS Input Column Audit

Purpose: Check whether a GC-MS table can enter uafR.

Required fields: file name; observed columns; required columns; missing columns; renamed columns; duplicate peak keys; suspicious match factors; suspicious areas; decision.

Completion standard: `spreadOut()` is run only after required columns and duplicate-key risks are inspected.

7.7 Query Settings Log

Purpose: Make database enrichment reproducible.

Required fields: date; function; compounds; spelling; cache setting; cache directory; throttle; profile/detail; query batch; network status; saved output path; validation output path; source coverage note.

Completion standard: A future student can rerun or inspect the same query without guessing settings.

7.8 Trait Evidence Audit

Purpose: Decide which chemical traits are safe to analyze.

Required fields: compound; trait key; source table; source field; evidence text or ID; extraction rule; confidence; allowed values; analysis use; caveat; include/exclude decision.

Completion standard: Every trait used in a matrix, figure, or model has an evidence row.

7.9 ML Feature Inventory

Purpose: Define target and predictors before modeling.

Required fields: target variable; prediction task; feature name; source; unit/encoding; transformation; available at prediction time; leakage risk; missingness; include/exclude decision.

Completion standard: The feature table is built only from approved inventory rows.

7.10 Train/Test Split Plan

Purpose: Decide what kind of generalization the model is testing.

Required fields: split type; grouping variable; date cutoff if used; training rows; testing rows; reason; leakage risks; baseline model; metrics.

Completion standard: The split design is chosen before model comparison.

7.11 Model Card

Purpose: Summarize model purpose, data, performance, and limitations.

Required fields: model name; target; task type; data version; rows included/excluded; features; split design; algorithm; baseline; metrics; diagnostic plots; limitations; acceptable use; prohibited use; next validation step.

Completion standard: The model can be interpreted without opening the R script first.

7.12 Advanced ML Review Gate

Purpose: Decide whether advanced ML outputs improve the analysis or only add complexity.

Use during: Week 6, after the beginner model card is complete.

Required fields: beginner prerequisite check; advanced script outputs reviewed; time-holdout result; leave-one-site-out result; feature-set sensitivity result; conductivity decision; context-feature decision; top permutation-importance signals; skipped model warnings; final model decision; prohibited claims; next validation need.

Common mistakes: choosing the highest-performing sensitivity model without leakage review; treating permutation importance as causation; ignoring site-holdout weakness; hiding skipped overfit models; using synthetic teaching results as environmental evidence.

Completion standard: A reviewer can see why the preferred model was chosen, why rejected feature sets were rejected, and what claims remain out of bounds.

7.13 Figure Planning Worksheet

Purpose: Plan figures around claims and evidence.

Required fields: figure number; purpose; variables; data source; groups; plot type; units; sample counts; missingness; caption claim; caveat; script path; output path.

Completion standard: Every final figure has a planned claim and caveat.

7.14 Figure Caption Checklist

Purpose: Prevent vague or overstated captions.

Required fields: figure number; variables named; units shown; sample count; split design if model output; missingness noted; color/group definitions; claim strength; limitation sentence.

Completion standard: Captions are understandable without the main text.

7.15 Literature and Evidence Synthesis

Purpose: Connect results to published or official context.

Required fields: citation; system; method; variables; main result; limitation; relevance to project; difference from project; usable sentence.

Completion standard: Literature is used to contextualize, not to claim proof beyond the student's data.

7.16 Claim-Evidence-Limitation Memo

Purpose: Convert results into defensible writing.

Required fields: proposed claim; evidence source; supporting figure/table; limitation; safer wording; follow-up needed.

Completion standard: Every major report claim has a limitation and source.

7.17 Peer Review Form

Purpose: Give useful feedback on project clarity and rigor.

Required fields: reviewer; artifact reviewed; strongest part; unclear part; data concern; model concern; claim concern; reproducibility concern; required revision; optional improvement.

Completion standard: Feedback points to specific files, figures, or statements.

7.18 Daily Research Log

Purpose: Preserve the record of work.

Required fields: date; goal; files opened; commands/scripts run; outputs created; errors; decisions; evidence consulted; interpretation limit; next action.

Completion standard: The log makes the day's work reconstructable.

7.19 Weekly Progress Review

Purpose: Convert daily activity into milestone progress.

Required fields: week; milestone; deliverables completed; files produced; review feedback; changes made; blockers; next week priorities.

Completion standard: The next week starts with a clear action list.

7.20 Final Package Audit

Purpose: Confirm the project can be archived and continued.

Required fields: README present; raw data inventory; processed data; scripts; results; figures; report; presentation; session info; model card; evidence map; continuation note; rerun status; unresolved issues.

Completion standard: A future student can rerun the project or understand why specific steps require restricted data.

8 Rubrics

Rubrics define what counts as acceptable research work in this program. Scores can be adapted, but evidence quality should remain central.

8.1 Weekly Deliverables Rubric

Criterion	Weight	Meets standard	Needs revision
Milestone completion	30%	Required products exist and match the week's decision gate.	Missing product, unclear scope, or artifact does not answer the week's question.
Evidence traceability	25%	Claims point to source files, tables, figures, or citations.	Claims are broad, unsupported, or source roles are vague.
Reproducibility	25%	Scripts rerun from project root and outputs are documented.	Hidden console objects, broken paths, or unclear script order.
Research log	20%	Daily decisions, errors, and corrections are recorded.	Log is a summary only and cannot reconstruct work.

8.2 Research Log Rubric

Level	Description	Evidence
Excellent	The log is a technical record that another researcher could follow.	Dates, goals, files, commands, errors, decisions, outputs, and next steps are specific.
Acceptable	The log records most work and decisions but misses some details.	Main scripts and outputs are named, but some errors or reasons are incomplete.
Needs revision	The log reads like a diary or task list.	Missing file names, commands, decisions, or evidence.

8.3 Code and Reproducibility Rubric

Criterion	Meets standard	Needs revision
Project structure	Raw, processed, results, figures, models, logs, scripts, and reports are separated.	Files are mixed or outputs overwrite raw data.
Script order	README states exact run order and expected outputs.	Scripts must be run by memory or console history.
Paths	Relative paths work from project root.	Absolute machine-specific paths are required.
Session record	R and package versions are recorded.	Software environment is unknown.

8.4 Water-Quality QA/QC Rubric

Criterion	Meets standard	Needs revision
Variable definitions	Units, methods, ranges, and caveats are recorded.	Variables are modeled before they are defined.
Flags	Suspicious values are flagged with rules and raw values retained.	Values are deleted or changed without trace.
Missingness	Missingness is summarized by variable and group.	Missing data are ignored.
Interpretation	QA/QC limits appear in results and discussion.	Results assume data are clean without evidence.

8.5 uafR Evidence Traceability Rubric

Criterion	Meets standard	Needs revision
Input audit	GC-MS or query inputs are documented and checked.	uafR is run on unclear or malformed input.
Validation	<code>validateCategorateResult()</code> or saved validation outputs are inspected.	Enrichment outputs are interpreted without validation.
Trait evidence	Matrix or trait features trace to <code>chemicalTraitEvidence()</code> or an evidence table.	Trait columns are used as black-box variables.
Claim strength	Tentative annotation and database presence are described cautiously.	Claims imply confirmed identity, exposure, toxicity, or mechanism.

8.6 Machine-Learning Rubric

Criterion	Meets standard	Needs revision
Target definition	Target is measured, relevant, and not circular.	Target is vague, derived from predictors, or unavailable.
Split design	Split matches intended generalization and leakage risks are addressed.	Random split is used without considering repeated site/date structure.
Baseline comparison	Candidate model is compared to a simple baseline.	Only the preferred model is reported.
Metrics	Metrics match regression or classification task and test set is used.	Training performance or inappropriate metrics drive conclusions.
Model card	Data, features, algorithm, metrics, and limitations are documented.	Model cannot be interpreted without verbal explanation.
Advanced extension	Blocked validation, feature-set sensitivity, permutation importance, and decision gates are interpreted only after the beginner ladder is complete.	Advanced outputs are used to claim superiority without leakage review or split-design discussion.

8.7 Figures and Captions Rubric

Criterion	Meets standard	Needs revision
Purpose	Figure answers a specific question.	Figure exists because the data could be plotted.
Technical clarity	Axes, units, groups, sample counts, and missingness are visible.	Labels or units are missing.
Caption	Caption states evidence and limitation.	Caption overclaims or merely repeats the title.
Reproducibility	Figure script and output path are recorded.	Figure cannot be regenerated.

8.8 Final Report and Presentation Rubric

Criterion	Meets standard	Needs revision
Question and context	Report clearly links the question to WiCCED water-quality context.	Context is generic or disconnected from analysis.
Methods	Data, QA/QC, uafR settings if used, ML split, metrics, and software are specific.	Methods omit enough detail that rerun is impossible.
Results	Results are tied to figures/tables and avoid unsupported claims.	Results mix interpretation, causation, and speculation.
Discussion	Limitations and next steps are concrete.	Discussion hides uncertainty or overgeneralizes.

Presentation	Central result, evidence, model validity, and limitations are clear.	Slides are crowded, claim-heavy, or disconnected from evidence.
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9 Mentor Guide

This guide is written for the co-advising structure around uafR, water quality, aquaculture/ecosystem context, and machine learning. It is intentionally practical. The goal is to keep students moving while protecting scientific rigor.

9.1 Pre-Program Setup Checklist

- Decide whether each student starts with real approved data, teaching data, or both.
- Prepare data-use restrictions for unpublished or partner-owned files.
- Confirm RStudio and uafR installation path.
- Confirm whether live database access is allowed for `categorate()` and `pubchemProfile()`.
- Place any approved GC-MS files in a raw-data folder and record their provenance.
- Prepare weekly review times with both co-advising perspectives represented.
- Make the dsDNA Core logo and DSU branding assets available if the manual is rendered with final visuals.

9.2 Daily Check-In Pattern

A productive daily check-in can be short:

- What product is due today?
- What file will change?
- What evidence is needed before interpretation?
- What could go wrong technically?
- What claim is currently not supported?

9.3 Weekly Review Questions

Week	Review focus
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1	Is the question feasible, WiCCED-relevant, and bounded by available evidence?
2	Can the project folder be understood and rerun from the README?
3	Are variables, units, QA/QC rules, and missingness documented before analysis?
4	Is uafR being used only where chemical evidence supports it, and are claims cautious?
5	Is the ML target valid, is the split defensible, and is there a baseline?
6	Are features frozen, leakage risks addressed, beginner ML understood, and advanced validation/sensitivity outputs interpreted cautiously if used?
7	Do figures and literature synthesis support exactly the claims being made?
8	Can the final package be archived, rerun, and continued by another student?

9.4 When to Slow Down

Slow the program if:

- The student cannot explain what one row of the dataset represents.
- Units, methods, or missing-value codes are unknown.
- uafR outputs are being interpreted before validation.
- Model performance is discussed before split design.
- Figures have claims that are stronger than the data.
- The research log is too thin to reconstruct work.

9.5 When to Accelerate

Accelerate only after the student has a reproducible pipeline and accurate interpretation habits. Strong students can add:

- The advanced ML extension, including site-holdout validation, feature-set sensitivity, permutation importance, and decision-gate review.
- A second target variable.
- Controlled chemical trait ontology matrices.
- Spatial covariates or remote-sensing context, with leakage review.
- A reviewer-response appendix.
- A short methods supplement suitable for a manuscript or symposium abstract.

9.6 Supporting Different Backgrounds

Students with weak R backgrounds should spend more time on Week 2 scripts, folder structure, and console interpretation. Avoid live web enrichment until local scripts run reliably.

Students with limited chemistry background should spend more time on Week 3 and Week 4 evidence types. Require written distinctions among measured variable, tentative annotation, database descriptor, and confirmed identity.

Students with strong data-science backgrounds should be challenged on leakage, sampling design, uncertainty, and scientific claim strength rather than simply adding more complex algorithms.

9.7 Real Data, Case Data, or No Data

The program works in three modes:

- **Real data mode:** Students use approved project data with explicit data-use notes.
- **Case data mode:** Students use de-identified or prepared project-like data.
- **Teaching data mode:** Students use the included synthetic teaching data to learn the workflow while real data are being prepared.

Teaching data should be labeled clearly as teaching data. It can support training products, but not environmental claims about Delaware waters.

9.8 Protecting Private and Unpublished Data

Project data may include unpublished sampling locations, partner information, or sensitive infrastructure context. Students should not email raw data, upload private data to public repositories, place unrestricted files in shared folders, or include exact sensitive locations in presentations unless approved. Public-facing outputs should use generalized site labels when required.

9.9 Final Package Evaluation

The final package is acceptable when:

- The README states the project question, data restrictions, script order, and expected outputs.
- Raw and processed data are separated.
- QA/QC decisions are documented.
- uafR evidence is validated and source-backed if used.
- ML models have baseline comparisons and model cards.
- Figures and report claims are consistent.
- A clean-session rerun succeeds or blocked steps are documented.
- The continuation note identifies next experiments and unresolved risks.

10 Program Completion Standard

Completion requires more than attendance. The final product must be auditable. A student has completed the program when the following artifacts exist and meet review standards.

10.1 Required Final Package

- Reproducible R project folder.
- README with project question, data restrictions, script order, and expected outputs.
- Raw-data inventory or teaching-data note.
- Cleaned water-quality data file.
- Water-quality variable dictionary.
- QA/QC flag log and missingness/source-coverage report.
- Project brief and scope/risk note.
- Optional uafR result object, validation output, source coverage, and trait evidence table if chemical screening is used.
- Not-applicable note if uafR chemical screening is not used.
- Analysis-ready ML feature table and feature dictionary.
- Train/test split plan and leakage audit.
- Baseline model and at least one candidate model or feature-set comparison.
- Model card with metrics, diagnostics, and limitations.
- Advanced validation, sensitivity, permutation-importance, and decision-gate outputs if the advanced ML extension is used.
- At least three publication-style figures or tables.
- Literature and evidence synthesis table.
- Manuscript-style report.
- Presentation or poster.
- Daily research log and weekly review notes.
- Session information.

- Final package audit.
- Continuation note with unresolved issues and recommended next steps.

10.2 Explicit Non-Claims

The final report must include a short statement of what should not be claimed. Depending on the project, this may include:

- The project did not confirm chemical identities without analytical standards.
- Public database annotations do not prove sample presence or biological activity.
- Pathway records do not demonstrate pathway activity in the water system.
- Hazard annotations do not constitute site-specific risk assessment.
- Machine-learning associations do not prove causation.
- The model is not ready for operational decision-making without external validation.
- Teaching data support training only, not environmental conclusions.

10.3 Final Acceptance Questions

Before archive, the project should answer yes to these questions:

- Can another student identify the research question from the README?
- Can the analysis be rerun from a clean R session?
- Are data restrictions clear?
- Are raw values preserved?
- Are excluded variables and rows explained?
- Are uafR outputs validated before interpretation?
- Are ML features frozen before final model comparison?
- If advanced ML was used, were blocked validation, sensitivity warnings, and decision gates reviewed before interpretation?
- Is the split design appropriate for the intended claim?
- Do figures state units, groups, sample counts, and limitations?
- Does the continuation note identify the most important next experiment?

A Appendix: Training Scripts and Data

The `training_wicced_water_quality/scripts` folder contains small scripts that support the manual. They are designed to run without live web services by default.

A.1 Script Inventory

Script	Purpose
<code>00_setup_check.R</code>	Reports R version, working directory, package availability, and uafR status.
<code>01_project_setup.R</code>	Creates a reproducible project skeleton with standard folders, README, and research log.
<code>02_water_quality_qaqc.R</code>	Reads the teaching water-quality data and writes QA/QC flags, cleaned data, and summaries.
<code>03_uafR_chemical_screening_template.R</code>	Runs the simulated WiCCED GC-MS hit-table workflow and writes spread, exact-match, abundance, context-summary, and evidence-note outputs.
<code>04_ml_01_build_feature_table.R</code>	Builds the beginner ML feature table, feature dictionary, split plan, training rows, and testing rows.
<code>04_ml_02_explore_features.R</code>	Writes feature summaries, correlations, class balance, exploration plots, and beginner exploration questions.
<code>04_ml_03_train_models.R</code>	Fits regression and classification baselines, writes metrics, predictions, coefficients, and a model-comparison summary.
<code>04_ml_04_diagnostics_and_model_card.R</code>	Writes diagnostic plots, diagnostic summary, interpretation guide, and model card.
<code>05_ml_01_advanced_validation.R</code>	Runs blocked validation after beginner ML is understood, including time-holdout and leave-one-site-out comparisons.
<code>05_ml_02_permutation_importance_and_sensitivity.R</code>	Tests feature-set sensitivity, conductivity sensitivity, skipped overfit feature sets, and held-out permutation importance.
<code>05_ml_03_advanced_model_report.R</code>	Writes the advanced model review, decision gate, and final recommendations.
<code>run_wicced_smoke_tests.R</code>	Runs setup, QA/QC, simulated GC-MS, the beginner ML ladder, the advanced ML extension, and expected-output checks in a temporary training project.

A.2 Teaching Data

`data/wicced_teaching_water_quality.csv` is a small synthetic teaching dataset. It is built to exercise QA/QC, plotting, and baseline modeling. It should not be used for environmental claims.

`data/chemical_screening_template.csv` is a small template for documenting candidate chemical screening records. It is not evidence that those compounds occur in a WiCCED sample.

`data/wicced_simulated_gcms_hits.csv` is a simulated uafR-compatible GC-MS hit table. It includes freshwater, agricultural drainage, urban stormwater, tidal-transition, marsh-edge, and aquaculture teaching samples. It is designed to demonstrate how uafR-style processing can organize candidate chemical evidence for water-quality research.

The simulated GC-MS pipeline is supported by three additional files in `data/`: a query list, a compound reference table, and sample metadata. These files are training scaffolds, not real field measurements.

A.3 Machine-Learning Resources

The `ml_resources/` folder contains the beginner ML guide, glossary, workflow checklist, troubleshooting notes, model-card field guide, and advanced ML extension guide. These resources are written for students who have not previously built a machine-learning model, with the advanced guide clearly marked as post-beginner work.

A.4 Running the Smoke Test

From the repository root:

```
Rscript training_wicced_water_quality/scripts/run_wicced_smoke_tests.R
```

The smoke test writes outputs to a temporary directory and reports the directory path. It checks whether the core scripts can run in sequence without live database access.

A.5 Live uafR Queries

Live `categorate()` or `pubchemProfile()` calls should be treated as optional enrichment. They require working package dependencies, internet access, respectful throttling, and careful validation. Students should start with one to three compounds and save the result object before scaling.

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